Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

A. 3.0 amps, 4 ohms

B. 2.8 amps, 3.2 ohms

C. 3.75 amps, 3.2 ohms

D. 4 amps, 3 ohms

E. 3.33 amps, 3.6 ohms

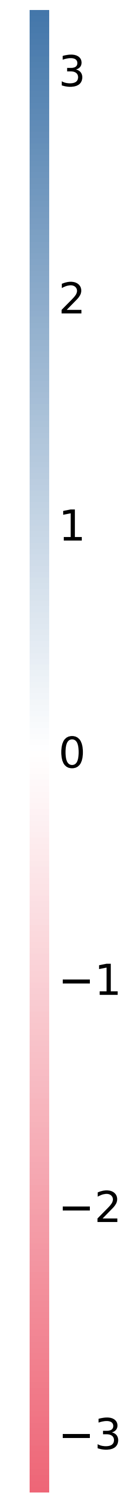
F. 5 amps, 2.4 ohms

G. 4.5 amps, 2.67 ohms

H. 2.2 amps, 5.45 ohms

I. 2.5 amps, 4.8 ohms

J. 3.5 amps, 4 ohms



Reasoning + Answer:

<think>

A battery with a 12 volt potential difference will try to drive 40 joules of heat through a resistive load. The electrical energy = power = work = 40 joules. The current flowing through a resistance R in ohms must be high enough to deliver 40 joules. Since energy (work) is current (rate of flow of charge) times voltage (work per unit of charge), then 40 joules = current (amps) x 12 volts. Rearranging this gives current = energy/voltage = 40 joules / 12 volts = 3.33 amps = 3.33 amps. This current flows through a resistance of volts per amp = 12 volts / current = 12 / 3.33 amps = 3.6 ohms.